

EXPERIMENT 4 – WORD FREQUENCY

Aim

To count the frequency of occurrence of a word in a body of text.

Procedure and Code

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg

nltk.download('gutenberg')
nltk.download('punkt')

sample = gutenberg.raw("austen-emma.txt")
token = word_tokenize(sample)

wlist = []
for i in range(50):
    wlist.append(token[i])

wordfreq = [wlist.count(w) for w in wlist]
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

OUTPUT

```
Pairs
[(['', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1),
('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma ', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',',
5), ('clever', 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1),
('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5), ('seemed',
1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 2), ('best', 1), ('blessings', 1),
('of', 2), ('existence', 1), (';', 1), ('and', 3), ('had', 1), ('lived', 1), ('nearly', 1),
('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2), ('world', 1), ('with', 2)]
```

Result

The frequency of occurrence of the first 50 tokens was successfully calculated.